

PARTICELLA (1D):

$$\hat{V} = 0$$

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}, \quad \hat{H}\mu = E\mu \Rightarrow -\frac{\hbar^2}{2m} \frac{\partial^2 \mu}{\partial x^2} = E\mu$$

$$\frac{\partial^2 \mu}{\partial x^2} = -\frac{2mE}{\hbar^2} \mu, \quad k^2 = \frac{2mE}{\hbar^2} \Rightarrow \frac{\partial^2 \mu}{\partial x^2} = -k^2 \mu \quad \left. \vphantom{\frac{\partial^2 \mu}{\partial x^2}} \right\} \begin{array}{l} E_q. \text{ oscillatore armonico} \\ \frac{d^2 x}{dt^2} = -\omega^2 x \\ \forall F: \text{fissato} \end{array}$$

$$\Rightarrow \mu(x) = A \cos(kx) + B \sin(kx) \quad k \in [0, +\infty]$$

ABBIA MO TROVATO UNA FAMIGLIA DI

SOLUZIONI

$$k_{\text{CONTINUO}} \Rightarrow E = \frac{\hbar^2 k^2}{2m} \quad \text{CONTINUA}$$

$$\mu(x) = A \frac{e^{ikx} + e^{-ikx}}{2} + B \frac{e^{ikx} - e^{-ikx}}{2i} = \underbrace{\left(\frac{A}{2} + \frac{B}{2i}\right)}_C e^{ikx} + \underbrace{\left(\frac{A}{2} - \frac{B}{2i}\right)}_D e^{-ikx} =$$

$$= C e^{ikx} + D e^{-ikx} \quad (\text{onda progr. + onda retr.})$$

$$\Rightarrow \Psi(x, t) = C e^{i(kx - \omega t)} + D e^{i(-kx - \omega t)} \quad \rightarrow \text{STATI STAZIONARI} \quad \omega = \frac{E}{\hbar}$$
$$\Psi = \mu(x) e^{-iE/\hbar t}$$

$$E = \frac{\hbar^2 k^2}{2m} \quad \text{Autovettore dell'op. } \hat{H}$$

Cosa possiamo dire di $\hat{p}$?

$$\hat{p}\mu_p = p\mu_p \quad \mu_p = C e^{ipx/\hbar}$$

$$p \in [-\infty, +\infty] \rightarrow \text{CONTINUO}$$

$\hookrightarrow$ Spettro continuo

$\rightarrow$ NORMALIZZAZIONE ALLA DIRAC: $\langle \mu_p | \mu_{p'} \rangle = \delta(p - p')$

$$\langle \mu_p | \mu_{p'} \rangle = \int_{\mathbb{R}} \mu_p^*(x) \mu_{p'}(x) dx = \int_{\mathbb{R}} C^* e^{ipx/\hbar} \cdot C e^{ip'x/\hbar} dx =$$

$$\int_{\mathbb{R}} |C|^2 e^{i(p' - p)x/\hbar} dx = \int (p - p')$$

$$|C|^2 2\pi\hbar \delta(p' - p) = \int (p - p') \quad \Rightarrow C = \frac{1}{\sqrt{2\pi\hbar}} \quad \Rightarrow \mu(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}$$

$$\hat{H} \Rightarrow \psi(x) = C e^{iKx} + D e^{-iKx} = C e^{i p x / \hbar} + D e^{-i p x / \hbar} \quad (\text{AUTOSTATI})$$

$$\hat{p} \Rightarrow \psi_p(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{i p x / \hbar}$$

$$\begin{aligned} &\text{Se } C=0 \text{ e } D \neq 0 \quad \Rightarrow \psi(x) = \psi_p(x) \\ &\text{Se } C \neq 0 \text{ e } D=0 \quad \Rightarrow \end{aligned}$$



$$[\hat{p}_x, \hat{H}] = [\hat{p}_x, \hat{K}] = [\hat{p}_x, K(p_x)] = 0 \Rightarrow \text{COMMUTANO} \Rightarrow \text{HANNO ALMENO 1 BASE IN COMUNE}$$

ESEMPIO:

$$\psi(x) = \frac{1}{\sqrt{3}} e^{iK_0 x} + \sqrt{\frac{2}{3}} e^{-iK_0 x}$$

$$E_0 = \frac{\hbar^2 K_0^2}{2m} \quad (\text{ben determin.})$$

$$\hat{p} \psi(x) = i\hbar \frac{\partial}{\partial x} \left[\frac{1}{\sqrt{3}} e^{iK_0 x} + \sqrt{\frac{2}{3}} e^{-iK_0 x} \right] = \underbrace{\hbar K_0}_p \left[\frac{1}{\sqrt{3}} e^{iK_0 x} - \sqrt{\frac{2}{3}} e^{-iK_0 x} \right] \neq \psi(x)$$

$$\psi(x) = \frac{1}{\sqrt{3}} e^{i p_0 x / \hbar} + \sqrt{\frac{2}{3}} e^{-i p_0 x / \hbar}$$

$$\Rightarrow \hat{p} \psi_p(x) = p_0 \psi_p(x) \Rightarrow \psi_p(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{i p_0 x / \hbar}$$

$$\Rightarrow \hat{p} \psi_{-p}(x) = (-p_0) \psi_{-p}(x) \Rightarrow \psi_{-p}(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{i (-p_0 x / \hbar)}$$

Autost. di $\hat{H}$

$$\psi(x) = \underbrace{1/\sqrt{3}}_a \psi_p(x) + \underbrace{\sqrt{2/3}}_b \psi_{-p}(x)$$

$$(\text{COMBINAZ. DEGLI AUTOSTATI di } p)$$

(p e $-p$ sono autoval.)

$$P(p=p_0) = \left| \frac{1}{\sqrt{3}} \right|^2 = \frac{1}{3}, \quad P(p=-p_0) = \frac{2}{3}$$

$$\langle \hat{p} \rangle = \frac{1}{3} (p_0) + \frac{2}{3} (-p_0) = -\frac{1}{3} p_0, \quad \langle \hat{p}^2 \rangle = \frac{1}{3} p_0^2 + \frac{2}{3} p_0^2 = p_0^2$$

$$\sigma_p = \Delta p = \sqrt{\langle \hat{p}^2 \rangle - \langle \hat{p} \rangle^2} = \sqrt{\frac{8}{9} p_0^2}$$

- MISURA $\Rightarrow$ TROVO $p_0 \Rightarrow$ COLLASSO $\psi_p(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{i p_0 x / \hbar}$

$$\langle \hat{H} \rangle = \frac{p_0^2}{2m}, \quad E_0 = \frac{p_0^2}{2m} \quad (\text{ben def})$$

$$\sigma_p = 0 \Rightarrow \langle \hat{p} \rangle = p_0$$

$\downarrow$
AUTOSTATO sia di $\hat{H}$ che $\hat{p}$

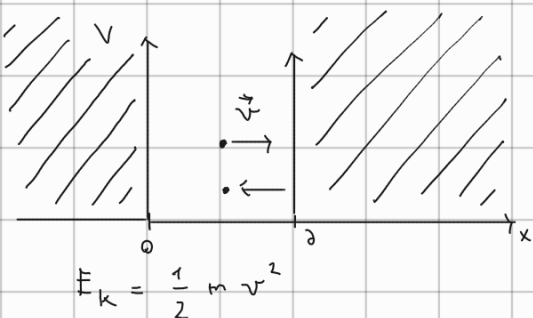
ONDA STAZ.: $\psi(x) = \frac{1}{\sqrt{2}} e^{i p_0/\hbar x} + \frac{1}{\sqrt{2}} e^{i (-p_0/\hbar) x} \Rightarrow \langle \hat{p} \rangle = 0$

$$P(p=p_0) = \frac{1}{2} \quad P(p=-p_0) = \frac{1}{2}$$

$$\Rightarrow -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = E \psi \Rightarrow \psi(x) = \int_{\mathbb{R}} A e^{i(K_0-K)x} dK \quad (\text{soluz. vera, pacchetto d'onde})$$

(A $\rightarrow$ A(K))

BUCA d: POTENZIALE INFINITA:



$$V(x) = \begin{cases} 0 & 0 \leq x \leq a \\ \infty & \text{ALTROVE} \end{cases}$$

① $-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V \psi = E \psi$

Se $x > a$ $V \rightarrow \infty$ MA $\psi \rightarrow \text{FINITA} \Rightarrow E \rightarrow \infty$

$\Rightarrow \psi(x > a, x < 0) = 0$

$\psi \rightarrow \text{CONTINUA} \Rightarrow \psi(x=0) = \psi(x=a) = 0$

② $0 \leq x \leq a$ $V=0 \rightarrow$ PARTICELLA LIBERA

$$E = \frac{\hbar^2 K^2}{2m} \quad K^2 = \frac{2mE}{\hbar^2} \Rightarrow \psi(x) = A \cos(Kx) + B \sin(Kx)$$

a) $\psi(x=0) = 0 \Rightarrow A=0$

b) $\psi(x=a) = 0 \Rightarrow B \sin(Ka) = 0 \Rightarrow Ka = n\pi \quad n=1,2,3,\dots$

$$\boxed{K_n = \frac{n\pi}{a}} \rightarrow K \text{ in una buca } \hat{=} \text{ QUANTIZZATO}$$

$$E_n = \frac{\hbar^2 K_n^2}{2m} = \frac{\hbar^2 \pi^2}{2ma^2} n^2 \Rightarrow E \text{ QUANTIZZATO}$$

$\hookrightarrow$ SPECTRO DISCRETO

$$\psi_n(x) = B \sin(K_n x) = B \sin\left(\frac{n\pi x}{a}\right)$$

NUMERO QUANTICO

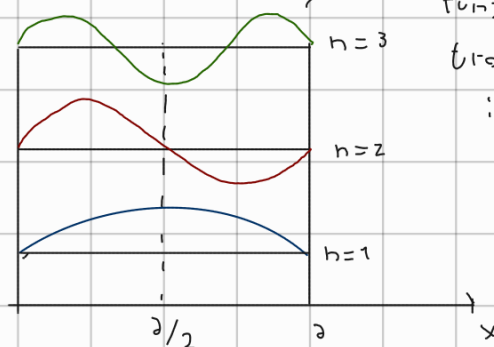
$$\rightarrow \int_0^a |\psi|^2 dx = 1 = \int_0^a |B|^2 \sin^2\left(\frac{n\pi x}{a}\right) dx \Rightarrow B = \sqrt{\frac{2}{a}}$$

$$\Rightarrow \boxed{E_n = \frac{\hbar^2 \pi^2 n^2}{2ma^2} \quad n=1, 2, 3, \dots}$$

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$$

$\Rightarrow$ AUTON. e AUTOE. di $\hat{H}$

E	$n=3$	$E_3 = \frac{9\hbar^2 \pi^2}{2ma^2}$
	$n=2$	$E_2 = \frac{4\hbar^2 \pi^2}{2ma^2}$
	$n=1$	$E_1 = \frac{\hbar^2 \pi^2}{2ma^2}$



Il $|\psi|^2$ di queste funz. è la P.d. trovare la prob. in x

- $n=1 \quad \psi_1(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right)$
- $n=2 \quad \psi_2(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{2\pi x}{a}\right)$
- $n=3 \quad \psi_3(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{3\pi x}{a}\right)$

① $n=0$ NON PERMESSO $E_0 = E_1 = \frac{\hbar^2 \pi^2}{2ma^2}$ se $n=0 \quad E=0 \Rightarrow \Delta p=0$
 $\downarrow$ En. di punto zero $\downarrow$ è fermo

Δx deve la massima valore $a \Rightarrow \Delta x \neq 0$

② $n=1 \quad \psi_1(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right) \rightarrow$ PARI (risq. ad $a/2$)

$n=2 \quad \psi_2(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{2\pi x}{a}\right) \rightarrow$ DISPARI

$n=3 \quad \psi_3(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{3\pi x}{a}\right) \rightarrow$ PARI

QUESTE AUTOE. HANNO PARITÀ BEN DEFINITA $\left(\begin{array}{l} n=\text{dispari} \Rightarrow \text{pari} \\ n=\text{pari} \Rightarrow \text{dispari} \end{array} \right)$

③ se $n \uparrow$ num. di nodi $\uparrow$

4) $\int_0^a \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) \cdot \sqrt{\frac{2}{a}} \sin\left(\frac{m\pi x}{a}\right) dx = \delta_{n,m} \rightarrow \text{ortogonali}$

5) $\Psi(x) = \sum_n c_n \psi_n(x) = \sqrt{\frac{2}{a}} \sum_n c_n \sin\left(\frac{n\pi x}{a}\right)$

$$c_n = \int_0^a \psi_n^*(x) \Psi dx = \sqrt{\frac{2}{a}} \int_0^a \sin\left(\frac{n\pi x}{a}\right) \Psi(x) dx$$

$$|c_n|^2 = P(E=E_n) \Rightarrow \sum_n |c_n|^2 = 1$$

6) Se CAMBIO gli ESTREMI della buca $\psi_n(x)$ cambiando



$$\psi(x = \pm \frac{a}{2}) = 0$$

7) $\Psi(x,t) = \sum_n c_n \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) e^{-i \frac{E_n}{\hbar} t}$

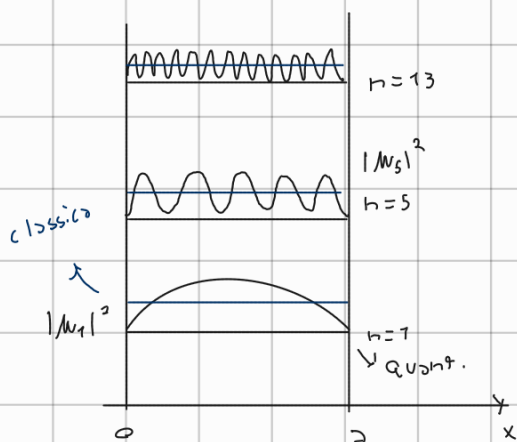
8) $L\hat{p} = 0 \Rightarrow$ autostati REALI, in media la v è 0

9) RACCORDO CON LIMITE CLASSICO:

$P_{cl}(x \div x+dx) \propto dt$, $dt = \frac{dx}{v} \Rightarrow \boxed{P_{cl} \propto \frac{1}{v}}$ $\rightarrow$ Per ogni stato E_k fissato $\Rightarrow v$ fissato

$\downarrow$
 $P_{cl} = \text{cost} = \frac{1}{v}$

$\propto$ Prob. classica (se posso più tempo tra x e $x+dx$ la prob. è alta)



$\Rightarrow$ più n cresce più la prob. class. e quant. si avvicinano

es. $m = 10^{-14} \text{ kg}$ (diam $\approx \mu\text{m}$), $a = 10 \text{ cm}$, $v = 1 \text{ mm/s}$

$$K = \frac{1}{2} m v^2 = 10^{-21} \text{ J} \quad \frac{\hbar^2 \pi^2}{2 m a^2} \approx 10^{-52} \text{ J} \quad \rightarrow \quad n = \sqrt{\frac{K}{\frac{\hbar^2 \pi^2}{2 m a^2}}} = 10^{15}$$

$$\delta E = E_{n+1} - E_n = \frac{\hbar^2 \pi^2}{2m a^2} [(n+1)^2 - n^2] \approx \frac{\hbar^2 \pi^2}{2m a^2} \cdot 2n \quad \int E \propto n$$

MA $E \propto n^2 \Rightarrow \frac{\delta E}{E} \propto \frac{1}{n}$

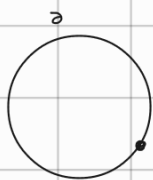
$$K_n = \frac{n\pi}{a} \rightarrow \text{gratto } K \text{ continuo} \quad \delta K = \frac{\pi}{a} \text{ LL } K = \frac{n\pi}{a} \quad n \approx 10^{15}$$



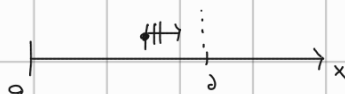
$$\int D(K) dK = \frac{dK}{\frac{\pi}{a}} = \frac{a}{\pi} dK$$

↓ # stati comp.
tro K ÷ K + dK

PARTICELLA CONFINATA SU UN ANELLO:



$$-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} = E \psi$$



$$\Rightarrow \psi(x) = A e^{iKx} \quad K \in [-\infty, +\infty]$$

Autostato di $\hat{H}$ e di $\hat{p}$

$$\int_0^a |\psi|^2 dx = 1 \Rightarrow |A|^2 a = 1 \Rightarrow A = \frac{1}{\sqrt{a}} \quad \Rightarrow \psi(x) = \frac{1}{\sqrt{a}} e^{iKx}$$

$$\psi(x+a) = \psi(x) \Rightarrow \frac{1}{\sqrt{a}} e^{iK(x+a)} = \frac{1}{\sqrt{a}} e^{iKx} \Rightarrow e^{iKa} = 1 \Rightarrow Ka = 2n\pi$$

Cond. di contorno
cicliche

$$\Rightarrow \boxed{K_n = \frac{2\pi n}{a}}$$

$$\int E_n = \frac{2\hbar^2 \pi^2}{m a^2} n^2$$

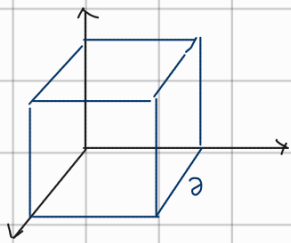
$$\rightarrow n=0 \quad \psi_0(x) = \frac{1}{\sqrt{a}} \quad E_0 = 0 \rightarrow \text{Non degenera}$$

$$\left. \begin{aligned} n = \pm 1 \\ \psi_1(x) = e^{i \frac{2\pi x}{a}} \\ \psi_{-1}(x) = e^{-i \frac{2\pi x}{a}} \end{aligned} \right\} E_1 = \frac{2\hbar^2 \pi^2}{m a^2} \quad \text{2 volte degenera}$$

$$n = \pm 2$$

$$n = \pm 3$$

BULCA di POTENZIALE INF. (3d)



$$V(x, y, z) = \begin{cases} 0 & 0 \leq x \leq a, 0 \leq y \leq a, 0 \leq z \leq a \\ \infty & \text{ALTROVE} \end{cases}$$

$$-\frac{\hbar^2}{2m} \Delta f(x, y, z) = E f(x, y, z)$$

→ SEPARO VARIABILI

$$f(x, y, z) = u(x) v(y) w(z)$$

$$-\frac{\hbar^2}{2m} \left[v w \frac{\partial^2 u}{\partial x^2} + u w \frac{\partial^2 v}{\partial y^2} + u v \frac{\partial^2 w}{\partial z^2} \right] = E u v w$$

$$-\frac{\hbar^2}{2m} \left[\underbrace{\frac{1}{u} \frac{\partial^2 u}{\partial x^2}}_{E_1} + \underbrace{\frac{1}{v} \frac{\partial^2 v}{\partial y^2}}_{E_2} + \underbrace{\frac{1}{w} \frac{\partial^2 w}{\partial z^2}}_{E_3} \right] = E \Rightarrow E = E_1 + E_2 + E_3$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 u}{\partial x^2} = E_1 u, \quad -\frac{\hbar^2}{2m} \frac{\partial^2 v}{\partial y^2} = E_2 v, \quad -\frac{\hbar^2}{2m} \frac{\partial^2 w}{\partial z^2} = E_3 w$$